



Load Short Form

Entire House

Job: 2141 Date:
FEBRUARY 2026
By: E

Project Information

For: JAY, PROFESSIONAL CAD DESIGN
PLAN SINGLE UNIT , PHOENIX,
AZ Phone: 623-249-1025

Design Information

	Htg	Clg	Infiltration	
Outside db (°F)	37	108	Method	Simplified
Inside db (°F)	70	75	Construction quality	Average
Design TD (°F)	33	33	Fireplaces	0
Daily range	-	M		
Inside humidity (%)	30	45		
Moisture difference (gr/lb)	7	-7		

HEATING EQUIPMENT

Make Rheem
Trade RHEEM
Model RJPL-A042JK000
AHRI ref TBD

Efficiency 8 HSPF
Heating input
Heating output 40000 Btuh @ 47°F
Temperature rise 34 °F
Actual air flow 1120 cfm
Air flow factor 0.095 cfm/Btuh
Static pressure 0.50 in H2O
Space thermostat
Capacity balance point = 10 °F

COOLING EQUIPMENT

Make Rheem
Trade RHEEM
Cond RJPL-A042JK000
Coil
AHRI ref TBD

Efficiency 11.6 EER, 14 SEER
Sensible cooling 31750 Btuh
Latent cooling 10250 Btuh
Total cooling 42000 Btuh
Actual air flow 1400 cfm
Air flow factor 0.052 cfm/Btuh
Static pressure 0.50 in H2O
Load sensible heat ratio 0.97

ROOM NAME	Area (ft²)	Htg load (Btuh)	Clg load (Btuh)	Htg AVF (cfm)	Clg AVF (cfm)
BEDROOM 1	117	1955	2654	185	139
BEDROOM 2	117	1343	2247	127	118
LAUNDRY	34	803	1216	76	64
BATH	52	501	370	48	19
LIVING ROOM	162	4522	12993	428	682
KITCHEN	214	2705	7198	256	378

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Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.



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...CO) PLAN SINGLE UNIT \JAY (BOSCO) PLAN SINGLE UNIT 696.rup Calc =

MJ8 Front Door faces: E

Entire House	696	11822	26679	1120	1400
Other equip loads		1569	1574		
Equip. @ 1.00 RSM			28253		
Latent cooling			984		
TOTALS	696	13391	29237	1120	1400

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Building Analysis

Entire House

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AZ Phone: 623-249-1025

Design Conditions

Location:

Phoenix, AZ, US
Elevation: 1112 ft
Latitude: 33 °N

Outdoor:

Dry bulb (°F)
Daily range (°F)
Wet bulb (°F)
Wind speed (mph)

Heating

37
-
-
15.0

Cooling

108
22 (M)
70
7.5

Indoor:

Indoor temperature (°F)
Design TD (°F)
Relative humidity (%)
Moisture difference (gr/lb)

Heating

70
33
30
7.1

Cooling

75
33
45
-7.3

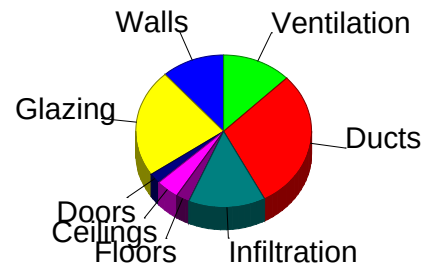
Infiltration:

Method
Construction quality
Fireplaces

Simplified
Average
0

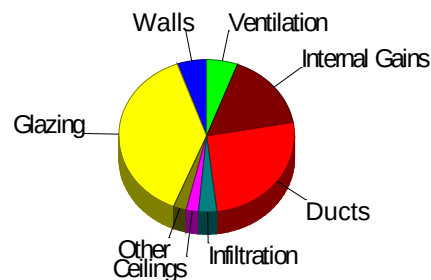
Heating

Component	Btuh/ft²	Btuh	% of load
Walls	2.2	1436	10.7
Glazing	14.6	2830	21.1
Doors	12.9	270	2.0
Ceilings	1.0	668	5.0
Floors	0.6	390	2.9
Infiltration	2.6	2221	16.6
Ducts		4007	29.9
Piping		0	0
Humidification		0	0
Ventilation		1569	11.7
Adjustments		0	0
Total		13391	100.0



Cooling

Component	Btuh/ft²	Btuh	% of load
Walls	2.3	1499	5.3
Glazing	54.6	10591	37.5
Doors	17.0	357	1.3
Ceilings	1.1	786	2.8
Floors	0.6	390	1.4
Infiltration	1.4	1168	4.1
Ducts		7348	26.0
Ventilation		1574	5.6
Internal gains		4540	16.1
Blower		0	0
Adjustments		0	0
Total		28253	100.0



Latent Cooling Load = 984 Btuh
Overall U-value = 0.082 Btuh/ft²-°F

WARNING: window to floor area ratio = 27.9% - more than 25%.

Bold/italic values have been manually overridden



Project Summary

Entire House

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For: JAY, PROFESSIONAL CAD DESIGN
PLAN SINGLE UNIT , PHOENIX,
AZ Phone: 623-249-1025

Notes:

Design Information

Weather: Phoenix, AZ, US

Winter Design Conditions

Outside db	37 °F
Inside db	70 °F
Design TD	33 °F

Summer Design Conditions

Outside db	108 °F
Inside db	75 °F
Design TD	33 °F
Daily range	M
Relative humidity	45 %
Moisture difference	-7 gr/lb

Heating Summary

Structure	7815 Btuh
Ducts	4007 Btuh
Central vent (45 cfm)	1569 Btuh
Outside air	
Humidification	0 Btuh
Piping	0 Btuh
Equipment load	13391 Btuh

Sensible Cooling Equipment Load Sizing

Structure	19331 Btuh
Ducts	7348 Btuh
Central vent (45 cfm)	1574 Btuh
Outside air	
Blower	0 Btuh
Use manufacturer's data	y
Rate/swing multiplier	1.00
Equipment sensible load	28253 Btuh

Infiltration

Method	Simplified
Construction quality	Average
Fireplaces	0

Latent Cooling Equipment Load Sizing

Structure	1341 Btuh
Ducts	-141 Btuh
Central vent (45 cfm)	-216 Btuh
Outside air	
Equipment latent load	984 Btuh

	Heating	Cooling
Area (ft²)	696	696
Volume (ft³)	6264	6264
Air changes/hour	0.61	0.32
Equiv. AVF (cfm)	64	33

Equipment Total Load (Sen+Lat)	29237 Btuh
Req. total capacity at 0.70 SHR	3.4 ton

Heating Equipment Summary

Make	Rheem
Trade	RHEEM
Model	RJPL-A042JK000
AHRI ref	TBD
Efficiency	8 HSPF
Heating input	
Heating output	40000 Btuh @ 47°F
Temperature rise	34 °F
Actual air flow	1120 cfm
Air flow factor	0.095 cfm/Btuh
Static pressure	0.50 in H2O
Space thermostat	
Capacity balance point = 10 °F	

Cooling Equipment Summary

Make	Rheem
Trade	RHEEM
Cond	RJPL-A042JK000
Coil	
AHRI ref	TBD
Efficiency	11.6 EER, 14 SEER
Sensible cooling	31750 Btuh
Latent cooling	10250 Btuh
Total cooling	42000 Btuh
Actual air flow	1400 cfm
Air flow factor	0.052 cfm/Btuh
Static pressure	0.50 in H2O
Load sensible heat ratio	0.97

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...CO) PLAN SINGLE UNIT \JAY (BOSCO) PLAN SINGLE UNIT .rup Calc = MJ8

Front Door faces: E



AED Assessment

Entire House

Job: 2141 Date:
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By: E

Project Information

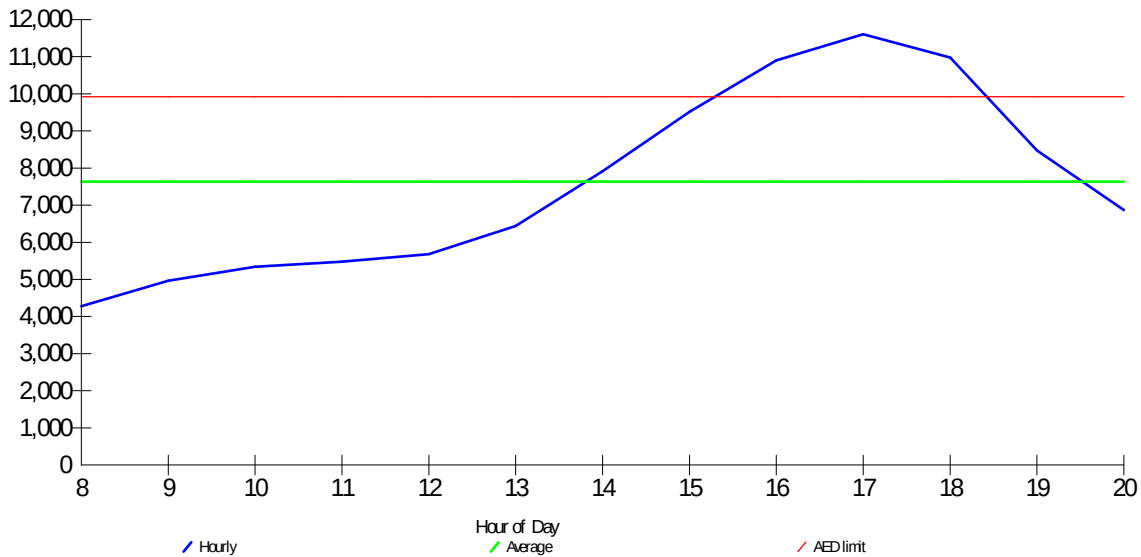
For: JAY, PROFESSIONAL CAD DESIGN
PLAN SINGLE UNIT , PHOENIX,
AZ Phone: 623-249-1025

Design Conditions

Location:		Indoor:		Heating	Cooling
Phoenix, AZ, US		Indoor temperature (°F)		70	75
Elevation: 1112 ft		Design TD (°F)		33	33
Latitude: 33 °N		Relative humidity (%)		30	45
Outdoor:		Infiltration:		7.1	-7.3
Heating	Cooling				
Dry bulb (°F)	37	Moisture difference (gr/lb)			
Daily range (°F)	-				
Wet bulb (°F)	-				
Wind speed (mph)	15.0				

Test for Adequate Exposure Diversity

Hourly Glazing Load



Maximum hourly glazing load exceeds average by 52.1%.

House does not have adequate exposure diversity (AED), based on AED limit of 30%.

AED excursion: 1687 Btuh (PFG - 1.3*AFG)

Bold/italic values have been manually overridden



Right-J® Worksheet

Entire House

Job: 2141

Date:

FEBRUARY 2026

By: E

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Calculations approved by ACCA to meet all requirements of Manual J 8th Ed.



Manual S Compliance Report

Entire House

Job: 2141 Date:
FEBRUARY 2026
By: E

Project Information

For: JAY, PROFESSIONAL CAD DESIGN
PLAN SINGLE UNIT PHOENIX,
AZ Phone: 623-249-1025

Cooling Equipment

Design Conditions

Outdoor design DB:	108 °F	Sensible gain:	28253	Btuh	Entering coil DB:	79.0 °F
Outdoor design WB:	69.9 °F	Latent gain:	984	Btuh	Entering coil WB:	62.1 °F
Indoor design DB:	75.0 °F	Total gain:	29237	Btuh		
Indoor RH:	45 %	Estimated airflow:	1400	cfm		

Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Pkg ASHP		
Manufacturer:	Rheem	Model:	RJPL-A042JK000
Actual airflow:	1400	cfm	
Sensible capacity:	33266	Btuh	118 % of load
Latent capacity:	1099	Btuh	112 % of load
Total capacity:	34364	Btuh	118 % of load SHR: 97 %

Heating Equipment

Design Conditions

Outdoor design DB:	37.0 °F	Heat loss:	13391	Btuh	Entering coil DB:	67.5 °F
Indoor design DB:	70.0 °F					

Manufacturer's Performance Data at Actual Design Conditions

Equipment type:	Pkg ASHP		
Manufacturer:	Rheem	Model:	RJPL-A042JK000
Actual airflow:	1120	cfm	
Output capacity:	34032	Btuh	254 % of load
Supplemental heat required:	0	Btuh	
Capacity balance:	10	°F	
Economic balance:	-99	°F	

REVIEW: total cooling capacity = 118% of load. ACCA Manual S limit is 115%.



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...CO) PLAN SINGLE UNIT \JAY (BOSCO) PLAN SINGLE UNIT .rup Calc = MJ8

Front Door faces: E



Static Pressure and Friction Rate

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AZ Phone: 623-249-1025

Available Static Pressure

	Heating (in H2O)	Cooling (in H2O)
External static pressure	0.50	0.50
Pressure losses		
Coil	0.19	0.19
Heat exchanger	0	0
Supply diffusers	0.04	0.04
Return grilles	0.04	0.04
Filter	0.11	0.11
Humidifier	0	0
Balancing damper	0	0
Other device	0	0
Available static pressure	0.12	0.12

Total Effective Length

	Supply (ft)	Return (ft)
Measured length of run-out	11	4
Measured length of trunk	7	0
Equivalent length of fittings	145	60
Total length	162	64
Total effective length		226

Friction Rate

	Heating (in/100ft)		Cooling (in/100ft)	
Supply Ducts	0.053	NOT OK	0.053	NOT OK
Return Ducts	0.053	NOT OK	0.053	NOT OK

Fitting Equivalent Length Details

Supply	4AD=60, 11T=25, 11T=25, 1A=35: TotalEL=145
Return	5D=40, 6M=20: TotalEL=60



Duct System Summary

Entire House

Job: 2141 Date:
FEBRUARY 2026
By:

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PLAN SINGLE UNIT , PHOENIX,
AZ Phone: 623-249-1025

	Heating	Cooling
External static pressure	0.50 in H ₂ O	0.50 in H ₂ O
Pressure losses	0.38 in H ₂ O	0.38 in H ₂ O
Available static pressure	0.12 in H ₂ O	0.12 in H ₂ O
Supply / return available pressure	0.086 / 0.034 in H ₂ O	0.086 / 0.034 in H ₂ O
Lowest friction rate	0.053 in/100ft	0.053 in/100ft
Actual air flow	1120 cfm	1400 cfm
Total effective length (TEL)	226 ft	

Supply Branch Detail Table

Name	Design (Btuh)	Htg (cfm)	Clg (cfm)	Design FR	Diam (in)	H x W (in)	Duct Matl	Actual Ln (ft)	Ftg.Eqv Ln (ft)	Trunk
BATH	h 501	48	19	0.055	6.0	0x0	VIFx	12.4	145.0	st3
BEDROOM 1	h 1955	185	139	0.053	10.0	0x0	VIFx	17.3	145.0	st1
BEDROOM 2	h 1343	127	118	0.056	10.0	0x0	VIFx	9.8	145.0	st1
KITCHEN	c 3599	128	189	0.085	10.0	0x0	VIFx	6.2	95.0	
KITCHEN-A	c 3599	128	189	0.086	10.0	0x0	VIFx	4.7	95.0	
LAUNDRY	h 803	76	64	0.054	7.0	0x0	VIFx	14.6	145.0	st3
LIVING ROOM	c 6497	214	341	0.069	12.0	0x0	VIFx	3.9	120.0	st2
LIVING ROOM-B	c 6496	214	341	0.067	12.0	0x0	VIFx	9.3	120.0	st2

Supply Trunk Detail Table

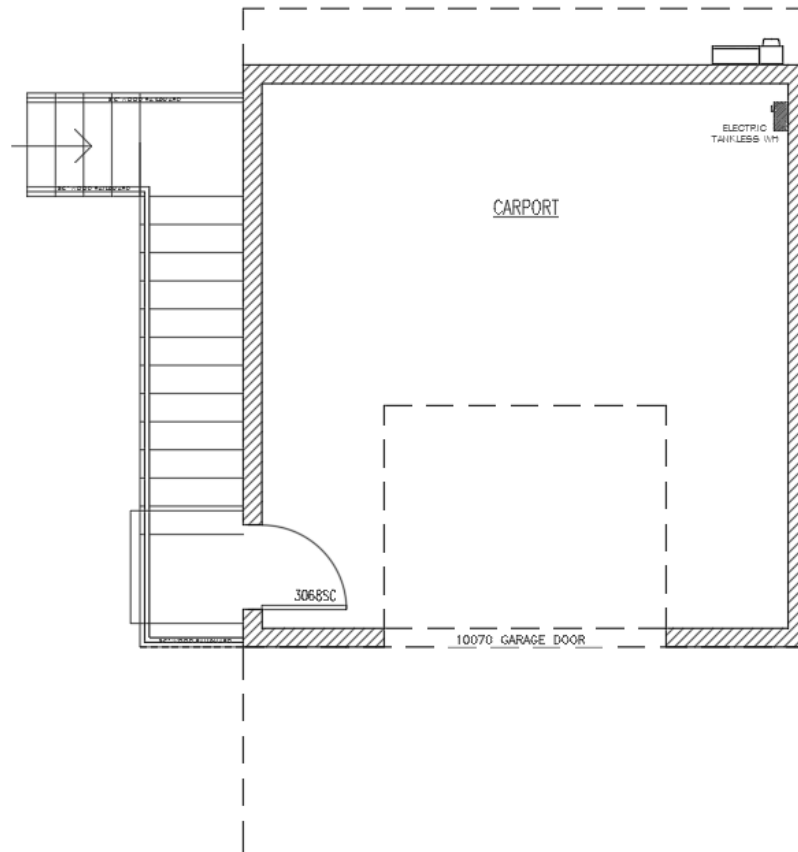
Name	Trunk Type	Htg (cfm)	Clg (cfm)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Duct Material	Trunk
st2	Peak AVF	428	682	0.066	489	16.0	0 x 0	VinIFlx	st1
st3	Peak AVF	124	83	0.054	227	10.0	0 x 0	VinIFlx	
st1	Peak AVF	436	340	0.053	408	14.0	0 x 0	VinIFlx	

Return Branch Detail Table

Name	Grille Size (in)	Htg (cfm)	Clg (cfm)	TEL (ft)	Design FR	Veloc (fpm)	Diam (in)	H x W (in)	Stud/Joist Opening (in)	Duct Matl	Trunk
rb1	0x0	1120	1400	63.8	0.053	530	22.0	0x 0		VIFx	



1ST FLOOR

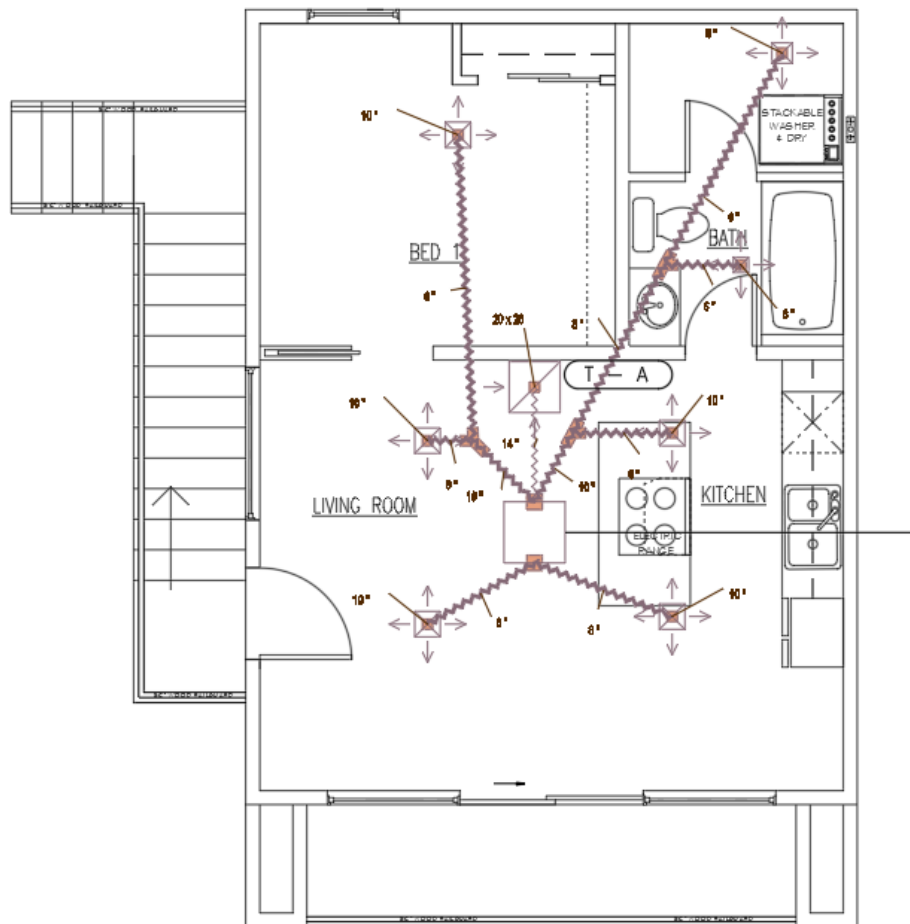


Job #: 2141
Performed by Evan for:
JAY
PLAN SINGLE UNIT
PHOENIX, AZ
Phone: 623-249-1025

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...\\JAY (BOSCO) PLAN SINGLE
UNIT .



2 ND STORY



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 PLAN SINGLE UNIT
 PHOENIX, AZ
 Phone: 623-249-1025

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